

Bluestock Mutual Fund Analytics & Quantitative Risk Platform

Technical Case Study & Executive Report

1. Platform Objectives & Background

This project establishes a unified analytics and risk-attribution engine for retail client portfolios and mutual fund performance. Historically, advisory teams and portfolio analysts have struggled with data fragmentation; Net Asset Values (NAV), retail transactional logs, and index benchmark feeds are scattered across separate systems and formats. To resolve this, we built a centralized data platform that ingests raw datasets, validates schema consistency, cleans duplicate or corrupt entries, and loads a dimensional STAR schema into SQLite. Furthermore, the engine runs quantitative models to analyze risk-adjusted performance, evaluates sector concentration, tracks retail investor vintage cohorts, and assesses tail-risk exposures to inform investment recommendations.

2. ETL Pipeline & Database Design

Our ETL pipeline is engineered in python (scripts/etl_pipeline.py). It reads raw CSV inputs, applies standardization rules, and loads a robust STAR schema. A key design choice was the strict normalization of the database schema (sql/schema.sql) to preserve referential integrity. The model consists of four dimension tables (dim_amc, dim_scheme, dim_date, and dim_investor) mapped to three fact tables (fact_nav_history, fact_investor_transactions, and fact_scheme_performance).

During implementation, we encountered and resolved several practical engineering challenges:

- **yfinance MultiIndex Resolution:** Downloading Nifty index data via the Yahoo Finance API returned MultiIndexed columns on single-ticker datasets. We implemented a programmatic column-flattening routine to prevent column shifts during pandas serialization.
- **Plotly Image Export Freezes:** Headless tasks on Windows often hang when using the Kaleido static engine. We engineered a custom Matplotlib/Seaborn visualization fallback that translates Plotly traces into high-fidelity static PNGs if Kaleido fails, while continuing to output interactive HTML files.
- **Anomalous Transaction Entries:** The raw transaction sheet contained 'INVALID' strings under transaction dates. We handled this by parsing dates with errors='coerce' and purging NaT rows to maintain database schema constraints.

Analytical Views are built into the database to drive Power BI dashboard indicators:

- **v_latest_nav:** Pulls the most recent Net Asset Value and daily change per scheme.
- **v_monthly_transactions:** Aggregates transaction counts, volume, and net investments grouped by month.
- **v_scheme_performance_latest:** Compiles the latest returns, AUM, and Sharpe snapshots.

3. Statistical Exploratory Data Analysis (EDA)

We generated 17 analytical charts to identify underlying relationships across returns, cost structures, and retail behavior. Our scheme correlation matrix (03_correlation_matrix.png) revealed high return correlations (above 0.85) between large-cap mutual funds, indicating that active managers tracking this segment closely mirror index holdings. In contrast, the ICICI Prudential Technology Fund exhibits a low correlation (~0.42) to the rest of the market, proving its high utility for diversification, though it introduces significant volatility. We also verified a clear cost drag on performance: plotting expense ratios against 3-year compound returns (10_expense_vs_returns.png) revealed a

negative relationship, confirming that higher fund fees systematically erode net compounding returns over time.

4. Performance Leaderboard & Scorecard

We created a weighted scorecard to rank the 10 mutual funds based on five factors: 3-Year Return (30% weight), Sharpe Ratio (25%), CAPM Alpha (20%), Expense Ratio (15% weight, inverse), and Maximum Drawdown (10% weight, inverse). This scorecard helps identify funds that generate superior returns relative to the risk they take.

Rank	Scheme Name	3Y CAGR	Sharpe	CAPM Beta	CAPM Alpha	Max DD	Expense
#1	DSP Top 100 Equity	39.78%	1.22	-0.04	28.88%	-24.73%	0.68%
#2	Mirae Asset Large Cap	33.15%	1.07	-0.03	23.87%	-17.47%	0.36%
#3	Axis Bluechip Fund	29.86%	0.96	0.02	21.23%	-23.78%	0.81%
#4	Canara Robeco Bluechip	4.72%	0.01	-0.03	0.28%	-24.11%	0.77%
#5	SBI Small Cap Fund	14.66%	0.41	0.05	9.17%	-31.97%	1.18%
#6	ICICI Pru Technology	2.13%	-0.09	0.04	-2.22%	-29.88%	1.16%
#7	Nippon India Large Cap	-2.15%	-0.24	0.06	-5.94%	-27.37%	0.76%
#8	Kotak Bluechip Fund	-2.60%	-0.28	-0.00	-6.40%	-42.49%	0.43%
#9	Parag Parikh Flexi Cap	-3.97%	-0.36	-0.09	-7.56%	-35.11%	0.34%
#10	HDFC Mid-Cap Opp.	-5.71%	-0.44	0.08	-10.18%	-49.77%	0.34%

5. Advanced Risk & Quantitative Analytics

Our advanced risk analysis evaluated portfolio concentration, downside tail risk, and investor behavior. Using the Herfindahl-Hirschman Index (HHI), we measured sector concentration. The ICICI Technology Fund exhibits a high HHI of 7,338.00 due to its focused exposure, while diversified funds show lower HHIs of ~1,827.16. This high sector concentration directly correlates with increased tail risk: the Technology Fund's daily Historical VaR (95%) is 2.33% and daily CVaR (95%) is 3.19%, compared to the Mirae Large Cap Fund's daily VaR of 2.08% and CVaR of 2.76%. Furthermore, our cohort analysis confirmed that investors utilizing Systematic Investment Plans (SIP) show higher retention rates (87.2% continuity rate) compared to lumpsum investors, indicating that systematic products encourage more disciplined investment behavior.

6. BI Dashboard Specifications

The dashboard uses a custom deep-dark theme (bluestock_theme.json) with charcoal card containers and electric cyan and green text to optimize readability. Page 1 (Industry Overview) displays KPI cards (Total AUM, SIP YTD, Active Folios, and SIP share) alongside inflow trends and AMC market share treemaps. Page 2 (Performance) visualizes risk-adjusted returns using a Beta vs. CAGR bubble chart, dynamic NAV growth comparison lines, and an interactive scorecard grid. Page 3 (Investor Analytics) features state-wise maps, transaction-type donut splits, and

age demographics, while Page 4 (SIP Trends) tracks monthly SIP volumes alongside the Nifty 50 Index.

7. Analyst Recommendations

- **Sector Concentration Controls:** Implement dynamic rebalancing protocols when a fund's HHI concentration exceeds 2,500 to mitigate sector-specific tail risk.
- **Promote SIP Products:** Emphasize SIP plans in retail marketing to build steady monthly inflows and improve investor holding periods.
- **Large Cap Anchoring:** Advisory teams should anchor conservative retail portfolios in Large Cap allocations to keep historical drawdowns above -20%.